WORKSHEET 8.7

Volumes from Known Cross-Sections

Unit 8: Applications of Integration

Key ideas

- If the cross-section at x has area $A(x)$, the volume is $\displaystyle\int_a^b A(x)\,dx$.
- The side s of each cross-section lies in the base: $s = \text{top} - \text{bottom}$.
- Square: s^2. Semicircle of diameter s: $\dfrac{\pi}{8}s^2$. Equilateral triangle: $\dfrac{\sqrt{3}}{4}s^2$.
- Isosceles right triangle with a leg in the base: $\dfrac{1}{2}s^2$. Rectangle of height h: sh.

Worked examples

Example 1 · Squares

The base of a solid is bounded by $y = x + 1$, the x-axis, $x = 0$ and $x = 3$. Cross-sections perpendicular to the x-axis are squares. Find the volume.

Solution. $\displaystyle\int_0^3 (x+1)^2\,dx = \frac{64-1}{3}$.

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Example 2 · Semicircles

The base is bounded by $y = \sqrt{x}$, the x-axis and $x = 9$. Cross-sections perpendicular to the x-axis are semicircles with diameter in the base. Find the volume.

Solution. $\dfrac{\pi}{8}\displaystyle\int_0^9 x\,dx = \frac{\pi}{8}\cdot\frac{81}{2}$.

$\dfrac{81\pi}{16}$.

Example 3 · Perpendicular to the y-axis

The base lies in the first quadrant, bounded by $y = x^2$, the y-axis and $y = 9$. Cross-sections perpendicular to the y-axis are squares. Find the volume.

Solution. The side is $x = \sqrt{y}$: $\displaystyle\int_0^9 y\,dy$.

$\dfrac{81}{2}$.

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Practice

CORE

1. Find the area of a semicircle with diameter 6.

2. Find the area of an equilateral triangle with side 4.

3. A solid has cross-sectional area $A(x) = 3x^2$ for $0 \leq x \leq 2$. Find its volume.

4. A solid has square cross-sections of side $2x$ for $0 \leq x \leq 3$. Find its volume.

EXAM LEVEL

5. The base of a solid is the disc $x^2 + y^2 \leq 25$. Cross-sections perpendicular to the x-axis are semicircles with diameter in the base. Find the volume.

6. The base is the region enclosed by $y = x$ and $y = x^2$. Cross-sections perpendicular to the x-axis are isosceles right triangles with a leg in the base. Find the volume.

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7. The base is bounded by $y = 6 - 2x$ and the coordinate axes. Cross-sections perpendicular to the x-axis are equilateral triangles. Find the volume.

8. *Calculator.* The base is bounded by $y = \ln x$, the x-axis and $x = e$. Cross-sections perpendicular to the x-axis are squares. Find the volume to three decimal places.

Question 9 · Free-response practice

Let R be the region bounded by $y = 6x - x^2$ and the x-axis. R is the base of a solid.

(a) Find the area of R.

(b) Cross-sections of the solid perpendicular to the x-axis are squares. Find the volume.

(c) For a different solid with the same base, the cross-section at x is a rectangle whose height is x. Find the volume.

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Worked solutions

1. $\frac{1}{2}\pi(3)^2 = \frac{9\pi}{2}$.

2. $\frac{\sqrt{3}}{4}(16) = 4\sqrt{3}$.

3. $\left[x^3\right]_0^2 = 8$.

4. $\int_0^3 4x^2\,dx = 36$.

5. Diameter $2\sqrt{25-x^2}$: $\frac{\pi}{8}\int_{-5}^{5} 4(25-x^2)\,dx = \frac{\pi}{2}\cdot\frac{500}{3} = \frac{250\pi}{3}$ (half of a sphere of radius 5).

6. $\frac{1}{2}\int_0^1 (x-x^2)^2\,dx = \frac{1}{2}\left(\frac{1}{3}-\frac{1}{2}+\frac{1}{5}\right) = \frac{1}{60}$.

7. $\frac{\sqrt{3}}{4}\int_0^3 (6-2x)^2\,dx = \frac{\sqrt{3}}{4}(36) = 9\sqrt{3}$.

8. $\int_1^e (\ln x)^2\,dx \approx 0.718$.

9. (a) $\int_0^6 (6x-x^2)\,dx = 108-72 = 36$.

   (b) $\int_0^6 (36x^2-12x^3+x^4)\,dx = 2592-3888+\frac{7776}{5} = \frac{1296}{5}$.

   (c) $\int_0^6 x(6x-x^2)\,dx = 432-324 = 108$.

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